This chapter lays the foundation for understanding digital identity before we explore authentication in chapter 3 and the basics of authorization in chapter 4. We will examine the challenges, properties, and models of digital identity, showing how they shape security and access control.

2.2 The problems of digital identity

The task of recognizing, remembering, and relating to others on the internet is not easy. Understanding the significant problems that digital identity systems face is important because it helps us understand why digital identity systems are built the way they are and where they still fail to completely meet the needs of many digital interactions.

2.2.1 The proximity problem

In the scenario at the start of this chapter, a close, human relationship was replaced by an experience intermediated by a computer. We all recognize the stark contrast in how the human and machine-based interactions feel because we've all been there. The kiosk may be located in the coffee shop, but it's just an interface for a system that's located somewhere else in a place we call "the cloud." The rich visual and conversational interaction with the barista has been replaced by a transactional experience based on a few specific pieces of information given to the underlying ordering system. This is the essence of the proximity problem.

Digital identity systems are far away from the entities they are creating relationships with and consequently rely on relatively weak signals. They manage the proximity problem by creating accounts to remember and use authentication mechanisms like usernames and passwords or biometrics to recognize the entities connecting to them. They also usually store attributes and other information that they need for context when interacting. These mechanisms introduce several problems, including user frustration and poor security. We'll talk more about these topics in the next chapter when we discuss authentication in detail.